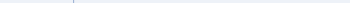
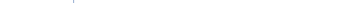
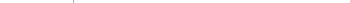




Summary

Sample	Spoligotype	Ocтал	Species	Lineage	Pattern (spacers 1 to 43)	Status
AF2122_97	SB0140	664073777777600	M. bovis	BOV		OK
H37Rv	Spoligo not found	777777477760771	M. tuberculosis	4.9		OK
NC_002945	SB0140	664073777777600	M. bovis	BOV		OK

Review

Spoligotypes are called from whole genome sequencing data. The evidence for each call (reads per spacer) is given in the sample sections, and the software, parameters and input files in the run information section.

Reviewed by		Date		Signature	
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AF2122_97 OK

Species and lineage

Region	Deleted in	Depth relative to MTBC control	Call
RD9	M. africanum and animal lineages (incl. M. bovis)	0.00	deleted
RD4	M. bovis and BCG	0.00	deleted
RD1	BCG (and M. microti)	0.80	present

Reads per spacer

Blue: present (count ≥ 5). Orange: called absent but seen in some reads.

H37Rv OK

[illegible]

Species and lineage

Species	M. tuberculosis
Lineage	4.9 · Euro-American (H37Rv-like) · typical spoligotypes: T1
MTBC DNA	median 1 contigs per control region, 100% of the control regions found

Region	Deleted in	Depth relative to MTBC control	Call
RD9	M. africanum and animal lineages (incl. M. bovis)	1.00	present
RD4	M. bovis and BCG	1.00	present
RD1	BCG (and M. microti)	1.00	present

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads
4	931,123	Rv0835	1	0
4.9	1,759,252	Rv1552	1	0

Contigs per spacer

Spacer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Count	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1
Spacer	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	
Count	1	1	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	1	1	1	

Blue: present (count ≥ 1). Orange: called absent but seen in some contigs.

NC_002945 OK

[illegible]

Run time	2.0 s
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Species and lineage

Species	M. bovis
Lineage	BOV · M. bovis · typical spoligotypes: BOV_2, BOV_1
MTBC DNA	median 1 contigs per control region, 100% of the control regions found

Region	Deleted in	Depth relative to MTBC control	Call
RD9	M. africanum and animal lineages (incl. M. bovis)	0.00	deleted
RD4	M. bovis and BCG	0.00	deleted
RD1	BCG (and M. microti)	1.00	present

Lineage SNP	Position (H37Rv)	Gene	Reads with lineage allele	Other reads
BOV_AFRI	1,882,180	Rv1662	1	0
BOV	2,831,482	Rv2515c	1	0

Contigs per spacer

Spacer	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Count	1	1	0	1	1	0	1	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1
Spacer	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	
Count	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	

Blue: present (count ≥ 1). Orange: called absent but seen in some contigs.

Run information

Operator	Example
User	example
Computer	example
Operating system	Linux-6.8.0-139-generic-x86_64-with-glibc2.39
Started	2026-09-24 13:15:00 UTC
Finished	2026-09-24 13:15:12 UTC
Duration	12.2 s
Working directory	/example
Command	spoligotyper -i data/ -o results/ -t 8 --operator Example

Parameters

Input	/example/data
Output folder	/example/results
Minimum count	default (5 for fastq, 1 for fasta)
Threads	8
Jobs	1
Seal memory	1g
Species and lineage	yes

Software

spoligotyper	0.4.1
Python	3.14.7 (/opt/conda/bin/python3.14)
Seal	/opt/conda/bin/seal.sh
BBTools	40.02
Java	openjdk version "25.0.2-internal" 2026-01-20

Reference data

Spoligotype database	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligotype_db.txt 1,976 patterns · MD5 0a88f7b89f746de2a11e1b4af0151e36
Spacer sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/spoligo_spacers.fasta 43 spacers · MD5 033c08e66aaa90193973f8d542e94c1
Species markers	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/markers.fasta MD5 55e4a9ef7d10c9a241b4a8ac4eb55200
Lineage SNP barcode	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_barcode.tsv MD5 28821eba812f42d2c0e9ce81053ae6cf
Lineage SNP sequences	/opt/conda/lib/python3.14/site-packages/spoligotyper/data/lineage_snps.fasta MD5 7e662a9baee450e79065705eb7b79021

Method

Spacers were counted with Seal (BBTools): each of the 43 spacers is searched as a single 25-mer on both strands, allowing 1 mismatch ($k=25$, $hdist=1$, $rcomp=t$, $maskmiddle=f$, $ambiguous=all$). A spacer is present when it is found in at least the minimum count of reads (contigs for assemblies). The binary pattern is converted to the octal code (Dale et al. 2001) and the hexadecimal code, and looked up in the spoligotype database for its SB number. Species: the read depth of regions of difference RD9, RD4 and RD1 (100 bp segments, same Seal parameters) is compared with the depth of MTBC-specific control regions; a region is deleted when its relative depth is at most 0.1, present when it is at least 0.5. The fraction of MTBC reads is the control depth divided by the depth expected from the number of bases. Lineage: reads carrying each allele of the 62 SNPs of the Coll et al. (2014) barcode are counted with exact 31-mers ($k=31$, $hdist=0$); a lineage is called when at least 80% of the reads (and at least 3, or 1 contig) carry its allele.

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